Multimodal Sensorized Feeding Nipple for Disease Detection in Calves

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Computer Science

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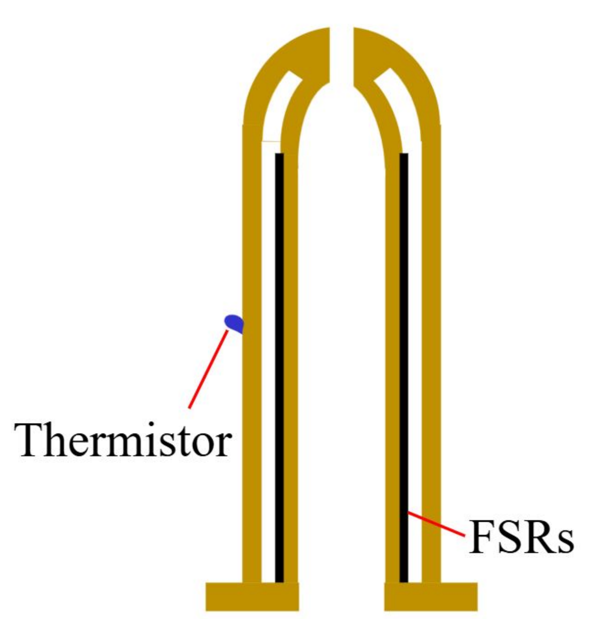
Introduction

Previous studies have shown that sick calves have a weaker suckling reflex compared to their healthy counterparts [1,2]. This work aims to build a multimodal sensor that can collect force and thermal data to quantify calf suckling behavior and aid in identification of sick calves.

The sensor is made of 12 force sensing resistors (FSRs) and a thermistor encased between two silicone layers. The electronics are sealed between the layers and protected from the liquid that flows through the center and from the outside environment.



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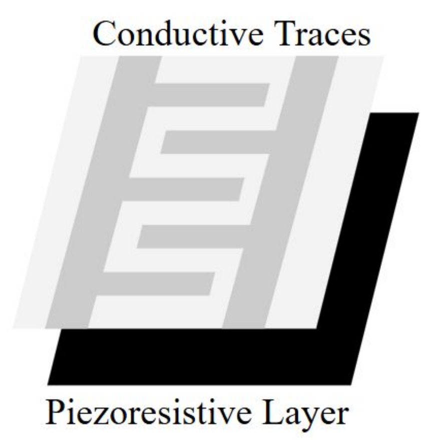
Sensor Construction

FSRs consist of a pressure sensing layer and a conducting layer. In a circuit, they act as a variable resistor. When it is pressed, the conductive layer touches the piezoresistive layer and the resistance goes down. The change in resistance can be calibrated to changes in force.

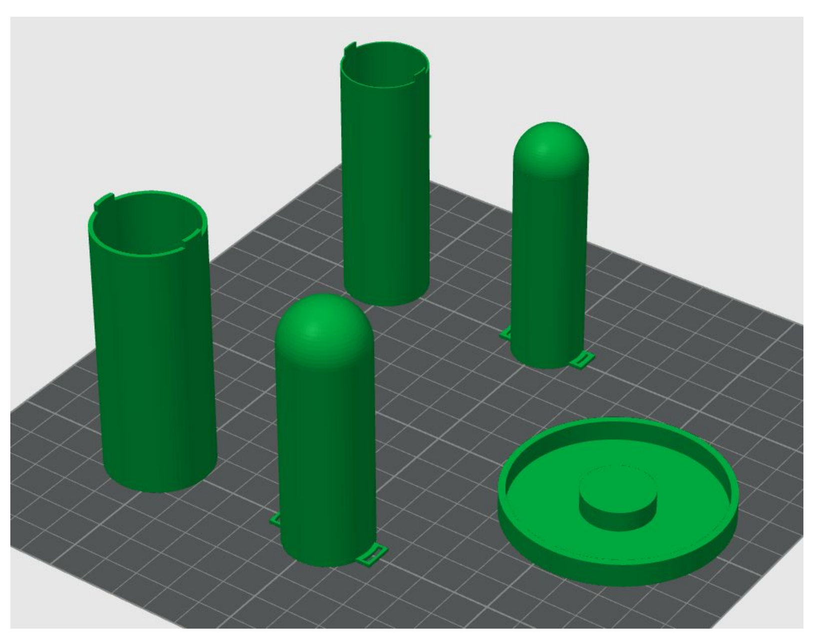


Similarly, thermistors change their resistance as temperature changes. The B parameter equation below is used to convert the measured resistance to temperature (K).

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \ln\left(\frac{R}{R_0}\right)$$

An Arduino Nano reads these changes through a voltage divider circuit.

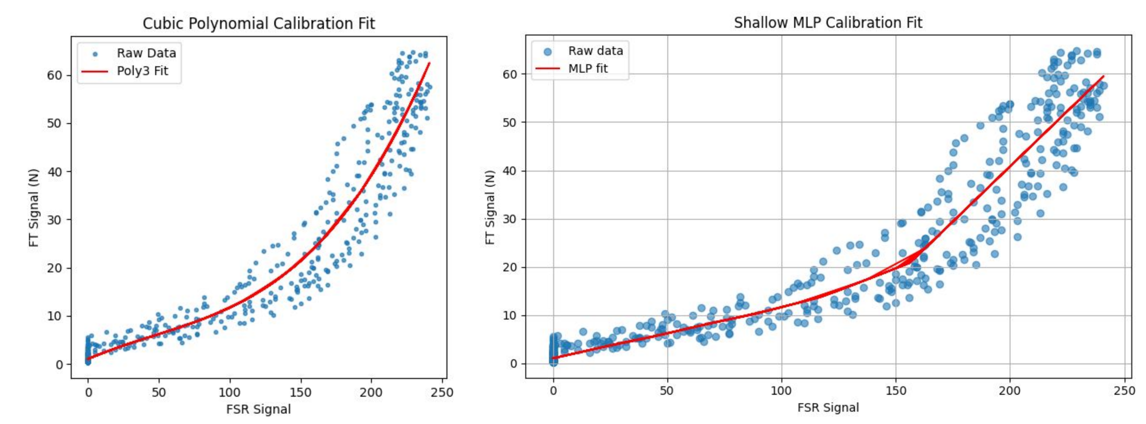
The nipple is made from a food safe silicone and shaped using 3D printed molds.



Calibration

To convert voltage readings to force in Newtons, we collect sensor voltage data and ground truth force data using an FT sensor to generate calibration functions.

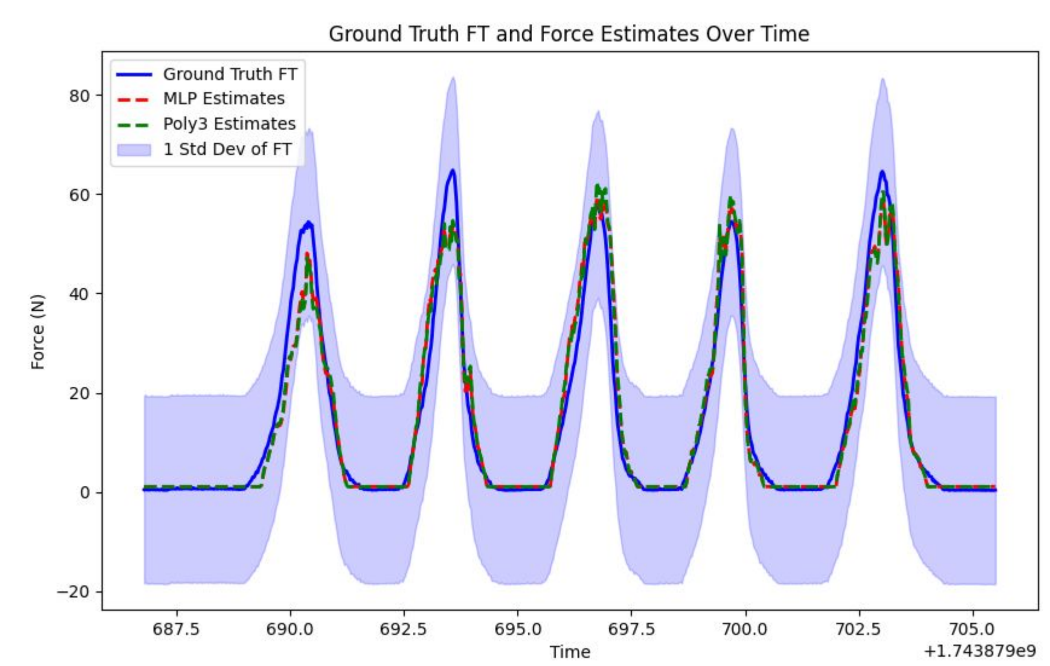
The cubic polynomial regression fits a third-degree polynomial to the data, while the shallow multilayer perceptron (MLP) approximates the voltage-to-force mapping by minimizing the error between predicted and true forces over multiple epochs.



Results

The sensor showed consistent performance across varying conditions.

This graph shows the calibration functions applied to sensor readings, demonstrating a good match to the ground truth force data within one standard deviation.



Conclusion

By continuously monitoring and analyzing a calf’s force suckling and thermal metrics, deviations from healthy baselines can signal potential health issues. This allows caretakers to take earlier action for faster interventions that improve calf welfare, reduce mortality rates, and lower treatment costs. This instrument provides real-time monitoring capabilities in a non-invasive, continuous manner for cost-efficient calf care.

Future Work

- Perform mechanical and electrical stress testing
- Explore more advanced ML algorithms for the calibration fit
- Conduct field testing with calves and collect live data
- Develop an ML algorithm to distinguish when a calf may be at higher risk of illness

References

[1] Chen, Z. M., von Konigslow, T. E., & Bhattacharjee, T. (2024). Development of Force Sensing Nipple for Describing Changes in Suckle Behavior Between Healthy & Sick Dairy Breed Calves. Paper presented at *ASABE Annual International Meeting July 28-31, 2024*. https://doi.org/10.13031/aim.202400903

[2] Boccardo, A., Biffani, S., Belloli, A., Biscarini, F., Sala, G., & Pravettoni, D. (2017). Risk factors associated with case fatality in 225 diarrhoeic calves: A retrospective study. *The Veterinary Journal*, *228*, 38–40. https://doi.org/10.1016/j.tvjl.2017.10.006